Mannan Khan

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EXPERIENCE

Viatron Technologies — **Suwon, South Korea**
Semiconductor Packaging Engineer (R&D) — *8/2024 – Present*

- Developed 3D models and detailed 2D engineering drawings for hybrid bonding equipment using Creo 3D and AutoCAD for semiconductor packaging.
- Designed and implemented motion control schemes for hybrid bonding and die attachment equipment.
- Programmed ACS motion controllers and drivers using ACSPL for advanced multi-axis motion control applications.
- Integrated and configured Servotronix linear and rotary motor drivers for high-precision motion planning.
- Performed system testing, validation and troubleshooting of hybrid bonder and die attachment equipment to ensure performance, stability, and manufacturing efficiency.

Korea Advanced Institute of Science & Technology (KAIST) — **Daejeon, South Korea**
Graduate Research Associate - Soft Robotic & Intelligent Materials (SRIM) Lab — *8/2022 – 7/2024*

- Conducted advanced research on smart wearable haptics systems incorporating shape memory alloys (SMA).
- Designed and executed experimental studies to improve tactile feedback performance in wearable devices.
- Collaborated with multidisciplinary research teams to develop innovative applications of smart wearable haptics technologies to improve user experiences.

ElectroCraft Engineering — **Lahore, Pakistan**
Assistant Manager Projects — *1/2022 - 6/2022*

- Led technical projects execution for Fire Detection & Alarm (Siemens), Lightning Protection (INGESCO), Earthing & Grounding (FURSE), and Public Address (ITC) systems, adhering to international standards.
- Designed, programmed, and implemented building automation projects using Siemens Desigo CC to client specifications and operational requirements.
- Conducted technical audits and reliability assessments to evaluate engineering solutions for compliance, safety, and performance standards.
- Coordinated with clients, vendors, and stakeholders to deliver cost-effective and technically optimized project solutions.

RapidTack — **London, United Kingdom**
Design Engineer — *8/2020 - 12/2020*

- Developed 3D CAD models, engineering simulations, and prototype designs in collaboration with a UK-based R&D engineering team.
- Led the design and simulation development of a Human-Assistive Exoskeleton system.
- Performed CAD-based analysis and iterative design improvements to optimize functionality, manufacturability, and system performance.

Nestle Sheikhupura Factory — **Lahore, Pakistan**
Associate Intern — *7/2019 - 8/2019*

- Conducted feasibility analysis to streamline machine breakdown entries on the production line.
- Implemented solutions to enhance efficiency and reduce redundancy in the incident reporting process.

EDUCATION

Korea Advanced Institute of Science & Technology (KAIST) — **Daejeon, South Korea**
MS in Mechanical Engineering — *8/2022 – 7/2024*

- Thesis topic: An Orthotropic Fabric Actuator for Wearable Haptics Based on Nested Auxetic-Architecture using Knotted SMA (Soft Actuators and their applications in Smart Wearables)
- CGPA: 3.66/4.3

University of Engineering & Technology (UET) — **Lahore, Pakistan**
BSc in Mechatronics & Control Engineering — *8/2016 - 7/2020*

- Thesis topic: Robot path planning and optimization using virtual reality
- CGPA: 3.53/4.0

SKILLS

- **Programming skills:** C, C++, Python, C#
- **AI & Computer Vision:** Machine Learning, OpenCV, Deep Learning
- **Robotics & Simulation:** ROS2, Gazebo, Unity 3D, Motion Planning, SLAM, Wearable Robotics
- **CAD & Engineering skills:** Matlab, SolidWorks, AutoCAD, Creo 3D, FEA, CFD, 3D printing
- **Circuit Design & PCB:** OrCAD Capture, PSpice, Proteus, Multisim, Altium
- **Hardware Control:** Arduino, Raspberry Pi, MyRio, PLC, Motion Controllers & Drivers (ACS, Servotronix, Eton, Delta, Hiwin, Siemens)
- **General skills:** Ms Office, Origin Pro
- **Languages:** Urdu (Native), English (Fluent), Korean (Beginner)

MAJOR PROJECTS

ROS2 & Gazebo Robotic Arm Simulation System

- Developed and simulated a robotic arm using ROS2 and Gazebo for robotic manipulation and motion control applications.
- Implemented joint control, motion planning, and real-time robot communication within the ROS2 framework.

LabVIEW-Based Shaking Table Control System

- Developed a LabVIEW-based control system for real-time shaking table motion and vibration testing.
- Integrated motion control and monitoring functionalities for precise dynamic system evaluation.

Hybrid Bonder & Die Attach UPH Optimization

- Improved Units Per Hours (UPH) for hybrid bonding and die attachment equipment through motion and process optimization.
- Optimized control sequences and machine parameters to improve production efficiency and operational stability.

Robot Path Planning & Optimization using Virtual Reality

- Developed a Unity 3D-based robotic path planning system for obstacle avoidance and trajectory optimization.
- Implemented real-world robot navigation using Arduino Mega 2560 and motion control algorithms.

Wearable EMG Monitoring System

- Designed and fabricated an EMG sensor system for human motion monitoring applications.
- Developed a MATLAB-based GUI for signal visualization and real-time data processing.

PLC-Based Industrial Automation Projects

- Developed PLC ladder logic systems for industrial process automation applications.
- Designed automated tank level control and traffic signal systems using PLC-based architectures.

Energy Harvesting from Walking

- Modelled and simulated an energy harvesting mechanism using MATLAB (Simulink) to generate electrical energy from walking motion.
- Evaluated system efficiency and energy conversion performance through simulation analysis.

Autonomous Line Navigation Robot

- Developed an autonomous mobile robot capable of real-time line tracking and obstacle avoidance using Arduino Mega 2560 and IR sensors.
- Implemented motion control and sensor integration algorithms for reliable autonomous navigation.

CAD Modeling Projects

- Designed mechanical assemblies including aircraft components, automotive systems, engine assemblies, and industrial machine parts using CAD and engineering analysis tools.

CERTIFICATION & TRAININGS

ACS Motion Controllers Training (PRESTO SOLUTION) **|** Python for Beginners (Coursera) **|** Advanced MATLAB (Mechatronics Club) **|** Siemens Fire Protection: Fire Alarm Systems - Level III **|** INGESCO Certified Lightning Protection System Installer (CLPSI) **|** ABB Furse Earthing solution

AWARDS & ACHIEVEMENTS

- Secure Marie Skłodowska-Curie Actions (MSCA), top European Fellowship (0.7% selection rate). (2025)
- Being a part of 7th International conference on Active Materials and Soft Mechatronics (AMSM 2024).
- Associated with 11th International Asian Conference on Multibody Dynamics 2024 (ACMD2024).
- Be a participant in Korean Society of Mechanical Engineering (KSME) conference (2024).
- Participated in Korean Society of Mechanical Engineering (KSME) conference (2023).
- Received two times Dean List of Honours in my Undergraduate (2020).
- Secured 3rd position in Design & Fabrication of Autonomous Vehicle competition held by MADE Foundation, MI, USA (2019).
- Runners up in LNR (Line Navigation Robot) Competition in UET Tech Week 2018.
- Won sketching & drawing competition at National Level (8th Grade).

RESEARCH PUBLICATIONS

- **M. Khan**, S. Oh, T.-E. Song, W. Ji, M. Mahato, Y. Yang, D. Saatchi, S. S. Ali, J. Roh, D. Yun, J.-H. Ryu, I.-K. Oh, Wearable Haptics for Orthotropic Actuation Based on Perpendicularly Nested Auxetic SMA Knotting. *Adv. Mater.* 2025, 37, 2411353. https://doi.org/10.1002/adma.202411353
- S. Oh, J. Jang, W. Ji, Y. Yang, **M. Khan**, C. Majidi, J.-H. Ryu, I.-K. Oh, Embodied Auxetic Intelligence in a Glove-Type Wearable Haptic Interface Connecting Humans to Robots and the Metaverse. *Adv. Funct. Mater.* 2025, 2502222. https://doi.org/10.1002/adfm.202502222
- B. Wicklein, H. Yoo, G. Valurouthu, J.-S. Kim, **M. Khan**, M. Mahato, F. Carosio, Y. Gogotsi, I.-K. Oh, Multifunctional Ti 3 C 2 T x-alginate foams for energy harvesting and fire warning. *Nanoscale Horiz.,* 2025, 10, 1084. https://pubs.rsc.org/en/content/articlelanding/2025/nh/d5nh00049a
- A. K. Taseer, S. Oh, J.-S. Kim, M. Garai, H. Yoo, V. H. Nguyen, Y. Yang, **M. Khan**, M. Mahato, I.-K. Oh, Cobalt MOF-Based Porous Carbonaceous Spheres for Multimodal Soft Actuator Exhibiting Intricate Biomimetic Motions. *Adv. Mater.* 2024, 36, 2312340. https://doi.org/10.1002/adma.202312340
- D. Saatchi, S. Oh, H. Yoo, J.-S. Kim, M.-J. Lee, **M. Khan**, B. Wicklein, M. Mahato, I.-K. Oh, Dynamic Schwarz Meta-Foams: Customizable Solutions for Environmental Noise Reduction. *Adv. Sci.* 2024, 11, 2402872. https://doi.org/10.1002/advs.202402872
- S. S. Ali, M. Mahato, D. V. Lam, P. Sambyal, G. Valurouthu, M. Garai, A. Dewan, V.H. Nguyen, **M. Khan**, A.K. Taseer, C. W. Ahn and I.-K. Oh., MXene-Cobalt Hybrid Electrodes for Electroactive Artificial Muscle. *Adv. Eng. Mater.* 2024, 26: 2400515. https://doi.org/10.1002/adem.202400515
- S. Oh, T.-E. Song, M. Mahato, J.-S. Kim, H. Yoo, M.-J. Lee, **M. Khan**, W.-H. Yeo, I.-K. Oh, Easy-To-Wear Auxetic SMA Knot-Architecture for Spatiotemporal and Multimodal Haptic Feedbacks. *Adv. Mater.* 2023, 35, 2304442. https://doi.org/10.1002/adma.202304442